

280-Gbps Rx-DSP-Free Coherent Analog-RoF Fronthaul Enabled by Hybrid Optical-Electrical Frequency Combs

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Abstract We demonstrate an Rx-DSP-free coherent A-RoF fronthaul using hybrid optical-electrical frequency combs to enable self-heterodyne detection and pre-equalization, mitigating laser phase noise and transmitter impairments. 5 channels of 14-GBaud 16-QAM signals with 22-GHz carrier frequency are successfully transmitted over a 10-km fiber link. ©2026 The Author(s)

Introduction

Driven by the burgeoning demands of IoT and ubiquitous streaming services, next-generation mobile networks impose unprecedented capacity requirements on the fronthaul. Meanwhile, the adoption of millimeter-wave and massive MIMO technologies necessitates the dense deployment of Remote Radio Units (RRUs) [1]. Consequently, simplifying the RRU architecture to mitigate the resulting surge in hardware costs and power consumption has emerged as a critical design imperative. In this context, Analog Radio-over-Fiber (A-RoF) presents a highly promising solution. By directly modulating analog RF signals onto optical carriers, A-RoF eliminates the need for energy-intensive digitization at the remote end [2]. This transparent signal forwarding enables a streamlined and cost-effective fronthaul network.

In pursuit of higher capacity, studies on A-RoF systems have moved towards elevated RF carrier frequencies, broader bandwidths and high-order modulation formats [3]. However, the realization of such high-capacity A-RoF systems is fundamentally constrained by signal fidelity degradation stemming from transmitter imperfections and laser phase noise [4]. Specifically, bandwidth limitation and IQ skew induced by RF circuit mismatches will lead to significant inter-symbol interference and severe channel crosstalk [5]. On the other hand, laser phase noise will cause random phase fluctuations in the recovered RF signal, resulting in constellation rotation and phase jitter. To combat these degradations, current high-capacity A-RoF solutions [6–9] rely heavily on complex receiver-side digital signal processing (DSP) for post-equalization. This reliance inherently reintroduces massive hardware and energy footprints at the remote units, undermining the low-complexity motivation of A-RoF architectures.

In this paper, we experimentally demonstrate an Rx-DSP-free coherent A-RoF fronthaul architec-

ture enabled by hybrid optical and electrical frequency combs (OFCs and EFCs), featuring a minimalist RRU design. The proposed scheme physically eliminates phase noise impairments through self-coherent beating of adjacent OFC lines, leveraging their intrinsic frequency locking and high mutual phase correlation. To simultaneously counteract signal fidelity degradation from transmitter imperfections, a comprehensive EFC-based calibration and pre-equalization technique is applied at the transmitter. Through this joint optical-electrical approach, analog RF signals are delivered to the RRU with high fidelity and directly upconverted to a higher intermediate frequency (IF) through self-heterodyne beating. Without any receiver-side post-equalization, we transparently forwarded 5 independent channels of 14-GBaud 16-QAM signals at 22 GHz after 10-km fronthaul fiber transmission, achieving an aggregate capacity of 280 Gbps (5 × 56 Gbps). This work validates a highly scalable and energy-efficient physical-layer solution for future 6G optical fronthaul systems.

Principle

Fig. 1(a) illustrates the experimental setup of the proposed Rx-DSP-free A-RoF system. At the distribution unit (DU), a continuous-wave optical seed laser is injected into cascaded intensity and phase modulators (IM and PM). Driven synchronously by an RF sinusoidal oscillator at ω_{RF} , these modulators generate an electro-optic frequency comb. The instantaneous optical phase of the k -th comb tone ($k = 0$ being the central comb line) is:

$$\Phi_k(t) = (\omega_c + k\omega_{RF})t + \theta_c(t) + k\theta_{RF}(t) \quad (1)$$

where $\theta_c(t)$ and $\theta_{RF}(t)$ represent the phase noise contributions from the seed laser and RF oscillator, respectively [10]. Eq. (1) indicates that the comb lines exhibit a frequency spacing strictly locked to ω_{RF} and highly correlated phase noise.

As shown in Fig. 1(b), an optical demultiplexer separates the comb lines into tone pairs for inde-

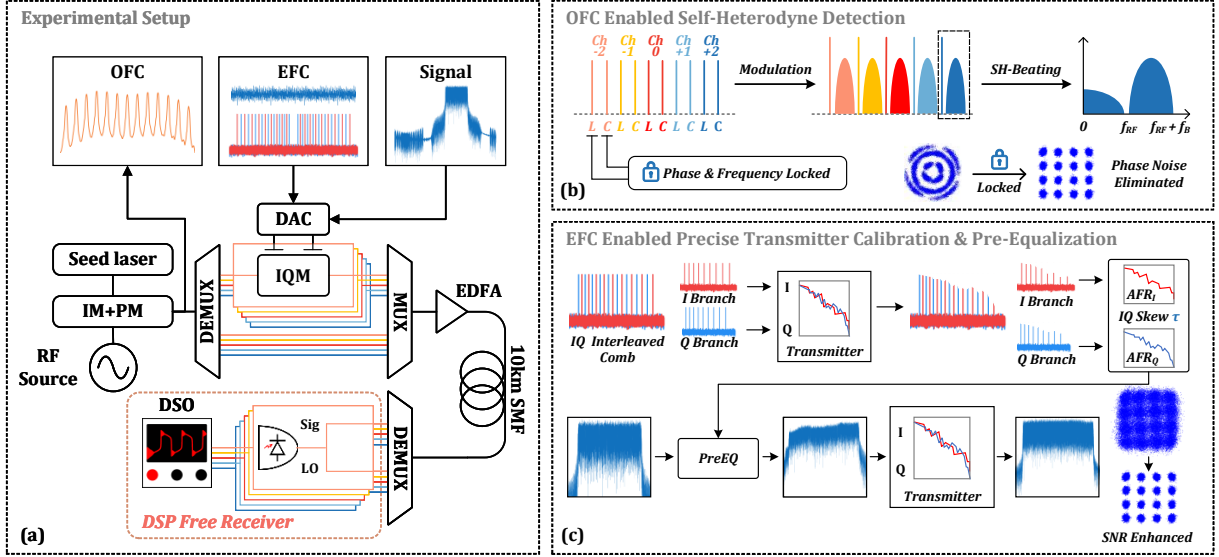


Fig. 1: (a) Experimental setup. (b) Schematic diagram of OFC-enabled self-heterodyne detection. (c) Schematic diagram of EFC-enabled transmitter calibration and pre-equalization

pendent channels, with one tone serving as the optical carrier and the adjacent tone as the local oscillator (LO). The data signals are modulated onto their respective carriers, combined with their corresponding LOs, optically amplified, and transmitted over a 10-km fronthaul fiber link. At the RRU, a photodetector (PD) performs self-heterodyne beating on each extracted pair. According to Eq. (1), this beating process inherently cancels the dominant laser phase noise $\theta_c(t)$, leaving only the negligible phase noise $\theta_{RF}(t)$ contributed by the RF oscillator.

Fig. 1(c) illustrates the calibration and pre-equalization procedure. A specially designed electrical frequency comb (EFC) is modulated onto the carrier to probe the channel response. The EFC tones for the I and Q channels are spectrally interleaved with linearly increasing frequency spacing, ensuring that their beat products are uniformly distributed without mutual overlap. The EFC is defined as $T_\eta(\omega) = \sum_n \delta[\omega - \omega_{\eta,n}] \exp(j\phi_{\eta,n})$, where $\eta \in \{I, Q\}$ denotes the branch index, and $\phi_{\eta,n}$ are predefined random phases introduced to reduce the peak-to-average power ratio (PAPR). Assuming the transmitter's channel transfer function is $H_\eta(\omega) = \alpha_\eta(\omega) \exp(j[\varphi_\eta(\omega) + \tau_\eta\omega])$, the output signal from the transmitter is:

$$S_\eta(\omega) = \sum_n \alpha_\eta(\omega_{\eta,n}) \times \exp(j[\varphi_\eta(\omega_{\eta,n}) + \phi_{\eta,n} + \tau_\eta\omega_{\eta,n}]) \quad (2)$$

After fiber transmission, the beating result between adjacent EFC tones on PD is:

$$R_\eta(\Delta\omega_{\eta,n}) = A_\eta(\Delta\omega_{\eta,n}) \times \exp(j\Phi_\eta(\Delta\omega_{\eta,n}))\delta(\omega - \Delta\omega_{\eta,n}) \quad (3)$$

which yields a new EFC in the lower-frequency range, where $\Delta\omega_{\eta,n} = \omega_{\eta,n} - \omega_{\eta,n-1}$. Assum-

ing the response of fiber and PD is flat across the calibration frequency range, the transmitter's amplitude frequency response (AFR) and phase frequency response (PFR) are embedded within the amplitude and phase of the newly generated EFC:

$$\begin{aligned} A_\eta(\Delta\omega_{\eta,n}) &= 2\sqrt{\alpha_\eta(\omega_{\eta,n})\alpha_\eta(\omega_{\eta,n-1})}, \\ P_\eta(\Delta\omega_{\eta,n}) &= \varphi_\eta(\omega_{\eta,n}) - \varphi_\eta(\omega_{\eta,n-1}) \\ &\quad + \phi_{\eta,n} - \phi_{\eta,n-1} \\ &\quad + \tau_\eta(\omega_{\eta,n} - \omega_{\eta,n-1}) \end{aligned} \quad (4)$$

By extracting the amplitude and phase of these EFC tones, the AFRs and PFRs of the I/Q branches can be reconstructed as:

$$\begin{aligned} \alpha_\eta\left(\frac{\omega_{\eta,n} + \omega_{\eta,n-1}}{2}\right) &= \frac{A_\eta(\Delta\omega_{\eta,n})}{2}, \\ \varphi_\eta(\omega_{\eta,n}) &= \sum_{k=1}^n [P_\eta(\Delta\omega_{\eta,k}) - \Delta\phi_{\eta,k} - \tau_\eta\Delta\omega_{\eta,k}], \end{aligned} \quad (5)$$

where $\Delta\phi_{\eta,k} = \phi_{\eta,k} - \phi_{\eta,k-1}$. Neglecting the PFR difference between the I and Q branches, IQ skew τ can be estimated by

$$\tau = d(\varphi_Q - \varphi_I) / d\omega \quad (6)$$

The DU then applies pre-equalization to mitigate the signal fidelity degradation induced by transmitter bandwidth limitation and IQ skew, and the compensated I and Q signals can be expressed as:

$$\begin{aligned} s_{I,EQ} &= \text{IFFT}\{\gamma\alpha_I^{-1}(\omega)S_I(\omega)\}, \\ s_{Q,EQ} &= \text{IFFT}\{\gamma\alpha_Q^{-1}(\omega)S_Q(\omega)\exp(j\omega\tau)\}, \end{aligned} \quad (7)$$

where $S_\eta(\omega)$ denotes the original signals, γ is a power scaling factor, and $\text{IFFT}\{\cdot\}$ denotes the inverse fast Fourier transform.

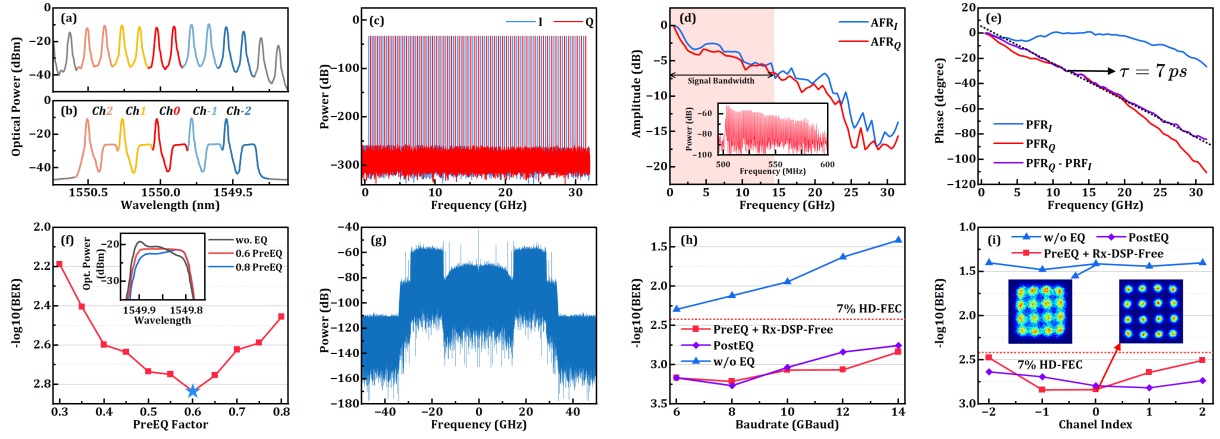


Fig. 2: (a) Optical spectra of the generated OFC. (b) Optical spectra of the transmitted signals. (c) Electrical spectrum of the EFC for Tx calibration. (d) Extracted AFR curves for the I and Q branches. (e) Extracted PFR curves for the I and Q branches. (f) Received electrical spectrum of Ch0 (14-GBaud 16-QAM at 22 GHz). (g) BER of Ch0 (14-GBaud 16-QAM) versus the PreEQ factor. (h) BER of Ch0 versus baud rate. (i) Multi-channel BER performance at 14-GBaud 16-QAM.

Experimental Results and Discussions

Empowered by the OFC-enabled self-heterodyne beating and EFC-enabled precise pre-equalization (PreEQ), five channels of 14-GBaud 16-QAM signals are successfully transmitted over a 10-km single-mode fiber (SMF). The 14-GBaud signals were digitally upconverted to an IF at 7 GHz and modulated onto 10 comb lines with a 15-GHz frequency spacing, achieving an aggregate fronthaul capacity of 280 Gbps. Fig. 2(a) and Fig. 2(b) depict the generated OFC and the optical spectra of the transmitted signals, respectively. The LO-to-signal power ratio is optimized to 15 dB to maximize the self-heterodyne beating efficiency.

Fig. 2(c) illustrates the specially designed EFC spectrum for transmitter calibration. The I and Q branches each contain 56 comb tones, covering a calibration bandwidth of 32 GHz. At the receiver, the beating of the adjacent EFC tones generates a new set of tones in the lower-frequency range. The beating result and the extracted AFRs of the I and Q branches are shown in Fig. 2(d). The signal suffers from more than 7-dB attenuation in the high-frequency domain due to bandwidth limitation. Furthermore, Fig. 2(e) presents the extracted PFR curves. According to Eq. (6), the IQ skew can be derived from the slope of the PFR difference; in this experiment, a 7-ps delay of the Q branch relative to the I branch was measured.

Having precisely calibrated the AFR and IQ skew, Tx-side PreEQ is applied based on Eq. (7) to mitigate the inherent high-frequency roll-off and IQ skew. To optimize the equalization performance, the PreEQ factor γ was swept. Fig. 2(f) shows the BER versus PreEQ factor for the central channel (Ch0) operating at 14-GBaud 16-QAM. Without PreEQ, severe high-frequency attenuation leads to significant signal fidelity degradation. As the PreEQ factor increases, the signal spectrum flattens and the BER steadily improves. However, an excessive PreEQ factor over-amplifies the high-

frequency components, causing the BER to rise again. An optimal PreEQ factor of 0.6 yields the minimum BER and the flattest spectrum within the signal bandwidth. The corresponding received signal spectrum is shown in Fig. 2(g), which shows decent signal integrity.

The system's transmission performance is further evaluated in Fig. 2(h) and Fig. 2(i) with a comparison between the proposed PreEQ scheme and the conventional post-equalization (PostEQ, which contains RDE and DD-LMS) scheme. As shown in Fig. 2(h), the BER of Ch0 deteriorates rapidly without equalization. By applying the optimal PreEQ factor of 0.6, the BER is significantly improved and remains below the 7% hard-decision forward error correction (HD-FEC) threshold (3.8×10^{-3}) for baud rates up to 14 GBaud. Across the entire range, the PreEQ scheme exhibits performance similar to that of PostEQ, showing the potential to eliminate Rx-side DSP. Fig. 2(i) presents the multi-channel performance at 14-GBaud 16-QAM. All five channels achieve BERs below the HD-FEC threshold, yielding an aggregate Rx-DSP-free fronthaul capacity of 280 Gbps.

Conclusions

We experimentally demonstrate a 280-Gbps Rx-DSP-free A-RoF fronthaul architecture enabled by hybrid optical and electrical frequency combs. This joint optical-electrical approach mitigates laser phase noise and transmitter impairments at the Tx side, ensuring high-fidelity transmission of the analog radio signal over the fronthaul link. The proposed scheme enables a minimalist RRU architecture free of energy-consuming digitization and post-equalization at the Rx side, presenting a highly scalable and efficient solution for next-generation mobile fronthauls.

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